

Rahika Jannat Roza

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Objective

Motivated Electrical Engineer with a passion for education, seeking opportunities to leverage academic excellence, leadership, and research skills for impactful roles.

EXPERIENCE

Research Internship | 6G Research Center | Khalifa University, Abu Dhabi | June 2025 – July 2025

- Replicated a system to recover images from electromagnetic emissions of HDMI cables (TEMPEST attack).
- Focused on digital signal reconstruction using Software Defined Radio (SDR) and GNU Radio framework.
- Addressed challenges of 10-bit encoded HDMI signals with non-linear pixel-intensity mappings.
- Framed the eavesdropping task as an inverse problem, applying deep learning to reconstruct displayed images.
- Designed a training methodology using mathematical signal analysis to generate synthetic data, avoiding the need for real-world capture setups initially.

Undergraduate Research | Khalifa University, Abu Dhabi | February 2025 – Present

- Conducted an in-depth literature review on signal processing in relay/cooperative system protocols.
- Analysed signal quality requirements to ensure adequate performance in IoT applications such as dispersed energy sources.
- Evaluated the physical layer security of cooperative wireless transmission systems using amplify-and-forward and decode-and-forward protocols.
- Determined QoS and security limits analytically and via simulations.
- Assessed challenges in different IoT network scenarios and prepared a comprehensive review of system capabilities and limitations.
- Currently working on publishing an Academic paper on this project .

Intern | ADNOC Offshore, Central Engineering Department | Summer 2024

- Collaborated with electrical engineering supervisors to enhance understanding of Single Line Diagrams and electrical distribution systems.
- Contributed to load scheduling tasks, ensuring accurate calculations of electrical loads and equipment sizing.
- Learned about Uninterruptible Power Supplies (UPS) and their role in maintaining power continuity in critical IT systems.
- Worked with switchgears, circuit breakers, and protective relays to ensure proper maintenance and safety compliance.
- Participated in discussions with cross-department supervisors including the Arzana Department and learned about P&IDs, drilling and heat exchangers

PROJECTS

Autonomous Indoor Drone-Based Asset Management System | Patent Pending | 2024 – Present

- Developed an autonomous drone-based system for warehouse asset detection, verification, and counting, integrating advanced machine learning techniques such as YOLO and MobileNet SSD for object detection and classification. The system also utilized RFID, barcodes, and QR codes for comprehensive asset identification.
- Designed and implemented a fully functional prototype, including hardware integration and software development.
- Applied deep learning models for real-time object detection and classification.
- Developed autonomous indoor navigation capabilities for efficient warehouse asset tracking.
- Conducted extensive system validation and performance assessment.
- Proposed improvements for enhanced detection accuracy and broader asset identification
- Presented at Innovation Day as the only undergraduate group selected
- Currently working on making this a start up company
- Presented at the IEEE conference and received First Place in the Senior Design Category
- Currently working on publishing an Academic paper on this project

Watermelon Ripeness Detection System Using YOLO, Audio Analysis, and Fusion Learning | Master’s Thesis Project

- Developed a non-destructive watermelon ripeness detection system using computer vision, acoustic signal processing, and fusion-based classification.
- Trained and evaluated YOLO-based object detection models to classify watermelon ripeness stages from images.
- Built an acoustic analysis pipeline using tapping sound recordings, FFT, PSD, resonant frequency, decay duration, and MFCC feature extraction.
- Implemented and compared audio classification models, including ANN, 1D-CNN, and 2D-CNN.
- Designed a fusion approach combining visual and acoustic predictions to improve ripeness classification reliability.
- Developed a mobile application prototype that can capture images, video, and audio for watermelon ripeness assessment.
- Integrated deep learning models with a backend API to support real-time prediction and application testing.
- Tools & Technologies: Python, YOLO, Roboflow, TensorFlow/Keras, FastAPI, React Native, Expo, Signal Processing, MFCC, FFT, PSD, Deep Learning

EDUCATION	Khalifa University, Abu Dhabi • MSc in Electrical and Computer Engineering • Expected Graduation: August 2026	Khalifa University, Abu Dhabi • Bachelor of Electrical Engineering • September 2021 – August 2025	Sabis International School, Ruwais • High School Degree • September 2008 – July 2021
SKILLS	• Public Speaker • MATLAB • Python • JavaScript • C ++ • Embedded systems development • Teaching	• Web development • ROS • Real-Time Operating Systems (RTOS) • Raspberry pi • Artificial Intelligence • Arduino • Soldering	

ACHIEVEMENTS	First Place Winner IEEE Conference Dubai University June 2025 • Presented our group’s Senior Design Project Autonomous Indoor Drone-Based Asset Management System • Explained the advanced AI-driven object detection, real-time classification, and autonomous navigation used in our project to the judges • Received positive feedback from the judges and helpful recommendations to improve our work Khalifa University Research and Innovation Exhibition 2025 • Selected as the only senior design startup to showcase at the highly competitive Khalifa University Research and Innovation Exhibition 2025. • Presented the Autonomous Indoor Drone-Based Asset Management System, demonstrating advanced AI-driven object detection, real-time classification, and autonomous navigation. • Engaged with industry experts, academics, and professionals, receiving recognition for innovation in warehouse asset management. • Highlighted groundbreaking work in deep learning and IoT-based asset tracking, setting a benchmark for future research and startups. Participant, Model United Nations (MUN) Khalifa University January 2024 • Engaged in diplomatic discussions and debates on global issues. • Developed public speaking, negotiation, and critical thinking skills. • Represented assigned country/committee and contributed to policy resolutions.
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EXTRACURRICULAR ACTIVITIES	• Participant, Mubarmij50 Robotics Competition January 2025. • Peer Tutor Khalifa University, Abu Dhabi January 2024 – May 2025 • Content Writer Khalifa University, Abu Dhabi January 2024 – May 2025 • Lead Peer Mentor Khalifa University, Abu Dhabi September 2023 – May 2025 • Academic Subhead Sabis International School, Ruwais September 2019 – July 2021 • Mentor Sabis International School, Ruwais September 2017 – July 2018 • Social Media Marketing Sabis International School, Ruwais September 2020 – July 2021
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